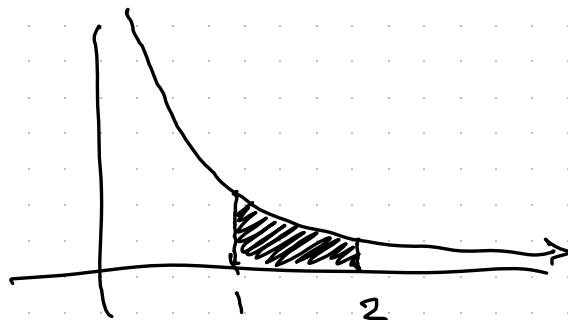


Improper integrals - 3.7

Example: Find $\int_1^2 \frac{1}{x^2} dx$

$$= -x^{-1} \Big|_1^2 = -\frac{1}{x} \Big|_1^2 = -\frac{1}{2} + \frac{1}{1} = \frac{1}{2}$$



The area under the curve between $x=1$ and $x=2$ is $\frac{1}{2}$.

What if we want to know the entire area under $\frac{1}{x^2}$ from 1 to ∞ ?

I.e. how to extend the definition of the integral to handle unbounded regions?

What about the area from 1 to N ?

$$\rightarrow \int_1^N \frac{1}{x^2} dx = -\frac{1}{x} \Big|_1^N = -\frac{1}{N} + 1 = 1 - \frac{1}{N}$$

Area of the entire infinite region bounded by $\frac{1}{x^2}$, the x -axis, and the line $x=1$ can be

described $\int_1^{\infty} \frac{1}{x^2} dx$ and computed as

$$\lim_{N \rightarrow \infty} \int_1^N \frac{1}{x^2} dx = \lim_{N \rightarrow \infty} 1 - \frac{1}{N} = 1$$

$$\int_1^{\infty} \frac{1}{x^2} dx = 1$$

Definition: The following are (type I) improper integrals.

$$- \int_a^{\infty} f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx$$

$$- \int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) dx$$

$$- \int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^{\infty} f(x) dx \quad \text{for any } c, \text{ often just choose } c=0$$

when these limits exist, we say that the integrals converge / are convergent. Otherwise, they diverge / are divergent.

$\int_{-\infty}^{\infty} f(x) dx$ converges iff both $\int_{-\infty}^c f(x) dx$ and $\int_c^{\infty} f(x) dx$ converge.

Example: $\int_1^{\infty} \frac{1}{x^2} dx$ is convergent.

Example: Does $\int_1^{\infty} \frac{1}{x} dx$ converge or diverge?

$$\rightarrow \int_1^{\infty} \frac{1}{x} dx = \lim_{b \rightarrow \infty} \int_1^b \frac{1}{x} dx = \lim_{b \rightarrow \infty} \ln|x| \Big|_1^b$$

$$= \lim_{b \rightarrow \infty} \ln(b) - \ln(1) = \lim_{b \rightarrow \infty} \ln(b) = \infty$$

Conclusion: $\int_1^{\infty} \frac{1}{x} dx$ diverges.

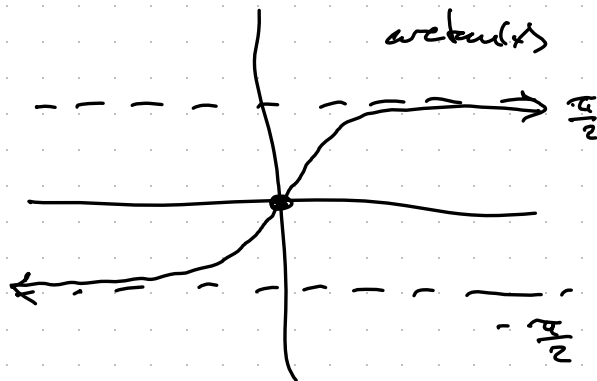
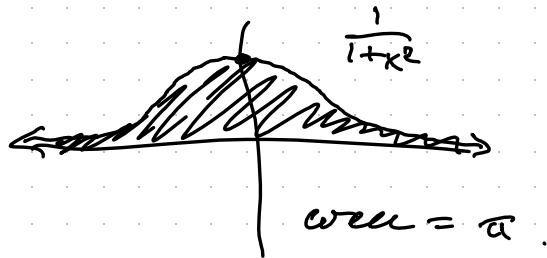
Example: Does $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$ converge or diverge?

$$= \int_{-\infty}^0 \frac{1}{1+x^2} dx + \int_0^{\infty} \frac{1}{1+x^2} dx = \lim_{a \rightarrow -\infty} \int_a^0 \frac{1}{1+x^2} dx + \lim_{b \rightarrow \infty} \int_0^b \frac{1}{1+x^2} dx$$

$$= \lim_{a \rightarrow -\infty} \arctan(0) - \arctan(a) + \lim_{b \rightarrow \infty} \arctan(b) - \arctan(0)$$

$$= \lim_{b \rightarrow \infty} \arctan(b) - \lim_{a \rightarrow -\infty} \arctan(a)$$

$$= \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi$$



Example: For which values of p does

$$\int_1^{\infty} \frac{1}{x^p} dx \quad \text{converge?}$$

Know that $p=1 \Rightarrow$ divergent

$p \leq 0 \Rightarrow$ divergent

$$\left(\begin{array}{l} \int_1^{\infty} \frac{1}{x^p} dx \text{ converges} \\ \text{if and only if } p > 1. \\ p \leq 1 \Rightarrow \text{diverges.} \end{array} \right.$$

$$\int_1^{\infty} \frac{1}{x^p} dx = \lim_{b \rightarrow \infty} \int_1^b \frac{1}{x^p} dx = \lim_{b \rightarrow \infty} \left. \frac{x^{-p+1}}{-p+1} \right|_1^b$$

$$= \lim_{b \rightarrow \infty} \frac{1}{-p+1} (b^{-p+1} - 1^{-p+1}) = \lim_{b \rightarrow \infty} \frac{1}{1-p} (b^{1-p} - 1)$$

$$\lim_{b \rightarrow \infty} b^{1-p} = \begin{cases} \infty & \text{if } 1-p > 0 \\ 0 & \text{if } 1-p < 0 \end{cases} = \begin{cases} \infty & p < 1 \\ 0 & p > 1 \end{cases}$$